Jacobian Analysis of a Recurrent Transformer Block

Dmitrii Vasilenko Ilia Stepanov Vadim Kasiuk

Gleb Karpeev Andrey Grabovoy

Moscow Institute of Physics and Technology

AINL 2026

Why Study Recurrent Transformers?

Looped (Recurrent) Architectures:

- Reuse the same block across iterations.
- Parameter-efficient: depth $\neq$ model size.

Key Question:
Do repeated applications of F converge stably, or does the gradient explode?

Iterative Map:

$$Y^{(t+1)} = F\left(Y^{(t)}\right)$$

$$F(Y) = \mathrm{RMSNorm}\big(Y + \nu\big(C + \mathrm{TB}(Y)\big)\big)$$

$\nu > 0$ — Step size coefficient
$C = Y^{(0)}$ — External context

The Role of the Jacobian

Why Jacobians?

- Classical tool for local stability analysis.
- If $\|J_F(Y)\|_2 < 1 \Rightarrow$ local contraction $\Rightarrow$ stability.

Prior Work (Tomihari & Karakida, 2025):

- Analyzed dynamics for *self-attention only* blocks.

Our Contribution:

We extend this framework to a full Transformer block including MSA, FFN, residual connections, and RMSNorm.

Architecture: Post-Norm Transformer Block

Notation: State $Y^{(t)} \in \mathbb{R}^{S \times D}$ with sequence length S, hidden dimension D.

Sub-blocks:

$$Y' = \mathrm{RMSNorm}(Y + \mathrm{MSA}(Y))$$

$$\mathrm{TB}(Y) = \mathrm{RMSNorm}\big(Y' + \mathrm{FFN}(Y')\big)$$

Full Iterative Update:

$$Y^{(t+1)} = \mathrm{RMSNorm}\Big(Y^{(t)} + \nu\big(C + \mathrm{TB}(Y^{(t)})\big)\Big)$$

*Row-wise RMSNorm defined as: $\mathrm{RMSNorm}(y) = \mathrm{diag}(\gamma)\,\frac{y}{\|y\|_2}$

Main Result: Spectral Norm Bound (Theorem 1)

Theorem 1. *Suppose inputs to all RMSNorm layers satisfy row-norm lower bounds:*

- $\|(Y + \mathrm{MSA}(Y))\| \geq r_{\mathrm{in}}$,
- $\|(Y' + \mathrm{FFN})\| \geq r_{\mathrm{out}}$,
- $\|(Y + \nu(C + \mathrm{TB}(Y)))\| \geq r_{\mathrm{U}}$,

where $r_{\mathrm{in}}, r_{\mathrm{out}}, r_U > 0$. *Let* $\gamma_{\max} = \max_j |\gamma_j|$. *Then the Jacobian spectral norm is bounded by:*

$$\|J_F(Y)\|_2 \leq \frac{\gamma_{\max}}{r_U}\left[1 + \nu\,\frac{\gamma_{\max}^2}{r_{\mathrm{out}}\, r_{\mathrm{in}}}\left(1 + \|J_{\mathrm{FFN}}\|\right)\left(1 + \|J_{\mathrm{MSA}}\|\right)\right]$$

Main Result: Asymptotic Stability (Theorem 2)

Theorem 2. *Under previous assumptions, and if* $\min_i \left\|\Delta Y_{[i,:]}\right\|_2 \geq \varepsilon > 0$*, the Jacobian norm remains bounded even for infinitely large step sizes* ν*:*

$$\|J_F(Y)\|_2 = O(1) \quad \text{as} \quad \nu \to \infty$$

Experiment 1: Validating Theoretical Bounds

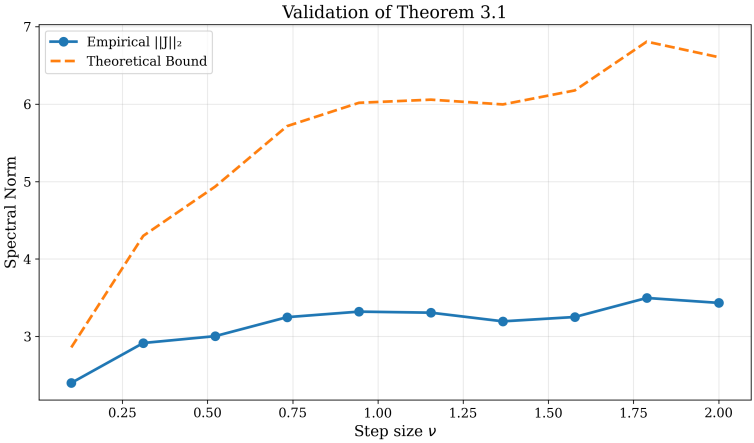

Figure: Empirical $\|J_F\|_2$ vs Theoretical Bound

Experiment 2: Validating Asymptotic Stability

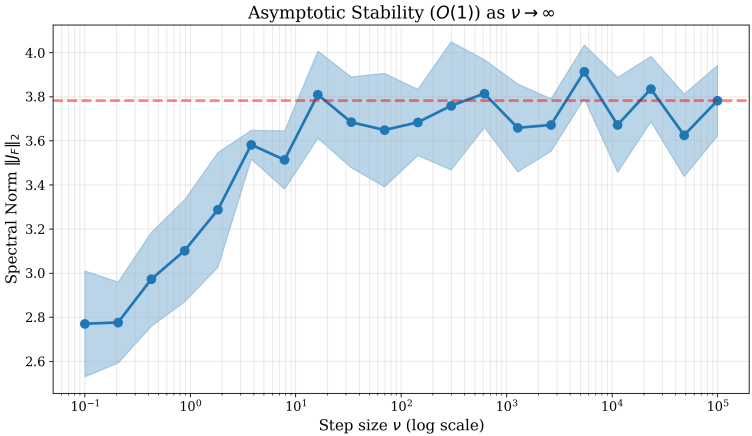

Figure: Asymptotic Stability as $\nu \to \infty$

Experiment 3: Contraction on Embeddings

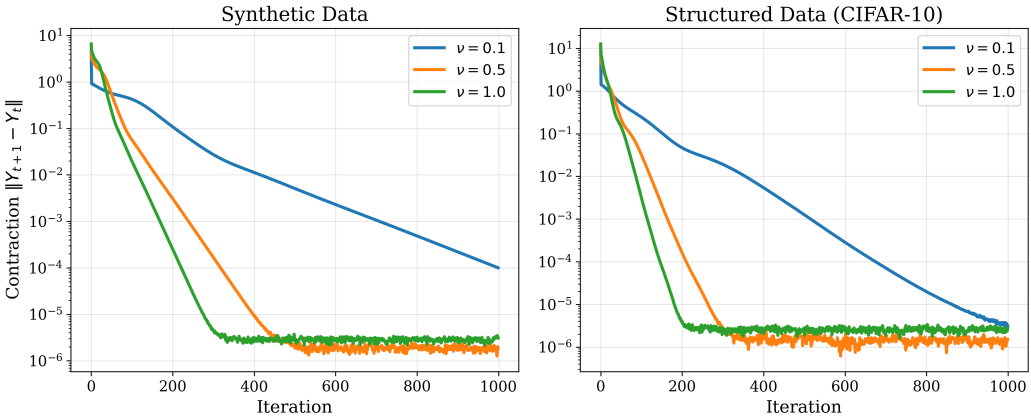

Figure: Contraction dynamics on synthetic data and real embeddings.

Conclusion & Future Work

Contributions:

- Jacobian analysis of a full Transformer block.
- Explicit spectral bounds derived.
- Proved $O(1)$ stability for large ν.

Future Directions:

- Tightening the spectral bounds.
- Extension to Pre-Norm architectures.

Thank you!

Paper, Slides & Code:

Click here

Dmitrii Vasilenko
vasilenko.dmitrii@phystech.edu